Assignment 5 — Data Preparation and Exploratory Data Analysis (EDA)

Structured thesis-oriented EDA report aligned with Assignments 3 and 4.

Spis treści

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Objective

This report documents a rigorous exploratory data analysis performed before formal preprocessing and modeling. The goal is not only to visualize the dataset, but to justify later engineering decisions in Assignment 6 and to verify whether the assumptions introduced in Assignments 3 and 4 remain defensible.

Data sources

The report relies primarily on the full raw GP demographics table stored in Excel format. It also references the currently available analytical table with the configured target variable, but only as a provisional check because that file is a very small reproducibility sample rather than the full thesis outcome dataset.

Dataset overview

• Raw demographics: 83,391 rows, 23 columns, date range 2014-01-01 to 2021-07-01, 1,082 practices, 16 Health Boards, 33 HSCP units.

• Analytical sample with target: 12 rows, 20 columns, date range 2024-01-01 to 2024-03-01, 4 practices.

• Field types: Date = reporting period; PracticeCode = practice-level panel key; HB / HSCP / Sex = categorical descriptors; AllAges + 8 age bands = numeric population counts; 9 QF columns = mostly missing quality flags that should not be treated as ordinary predictors.

A. Dataset Structure and Integrity Analysis

The raw table shows strong internal consistency. No full-row duplicates were detected, no duplicate Date–PracticeCode–Sex keys were found, and age-band arithmetic is coherent for almost all aggregate records.

The main data-quality issue is concentrated in the QF metadata columns. Their mean missingness is roughly 95.49%, whereas the worst missingness among the substantive non-QF fields is only 0.144%. This means the demographic counts are suitable for analysis, but QF columns should be excluded from ordinary quantitative modeling.

A few structural irregularities remain: 9 practice-date groups have an incomplete sex bundle, 0 complete triplets violate All = Female + Male, and 2 aggregate rows have AllAges <= 0. These cases are rare enough to handle explicitly during preprocessing.

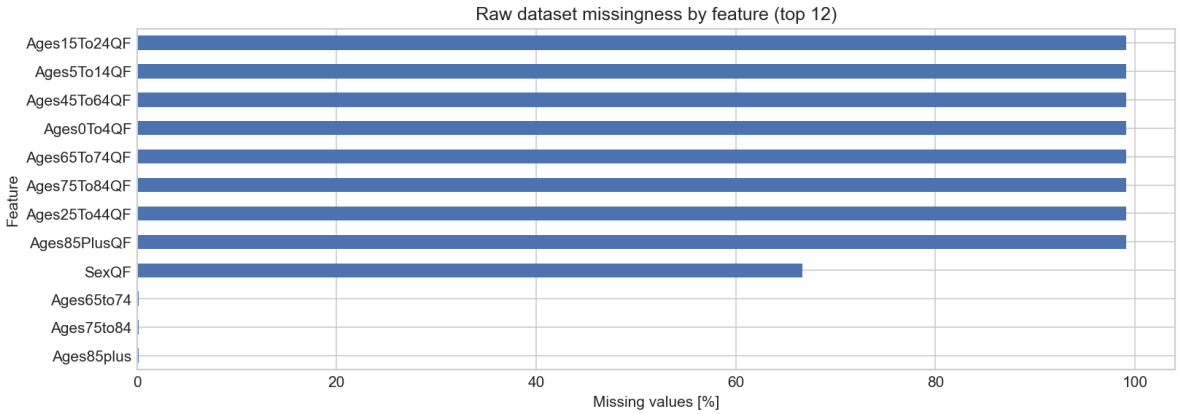

Figure 1. Missingness is concentrated in QF metadata fields rather than in the substantive demographic variables.

B. Statistical Characterization

The categorical structure is interpretable rather than pathological. The dataset contains 16 Health Boards and 33 HSCP units, while the largest HB contributes 19.16% of rows. This supports the use of regional controls, but it also warns against assuming equal support across administrative units.

The raw row distribution is structurally balanced by sex because the table is organized around the reporting convention All / Female / Male. That balance should not be confused with inferential balance for the final modeling problem.

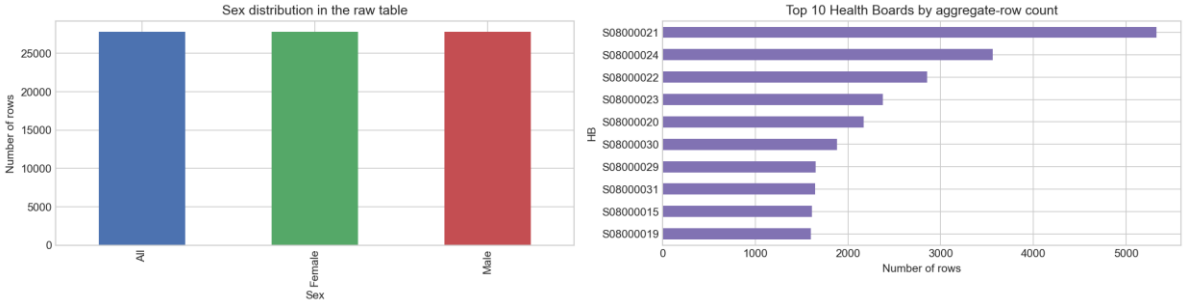

Figure 2. Categorical structure is stable and interpretable, but regional representation is uneven across Health Boards.

B. Statistical Characterization — Practice Size and Target Structure

The aggregate practice-size variable AllAges is clearly right-skewed. Its mean is 5,931.2, its median is 5,494.5, the skewness is 1.545, and the 99th percentile is about 17,557 patients. This indicates a long upper tail driven by a small number of very large practices.

The configured target remains a continuous regression variable, hospital_use_per_1000. In the repository sample it ranges from 244.1 to 347.9 with coefficient of variation 0.134, but that sample is far too small for substantive inference and should be treated as a pipeline check only.

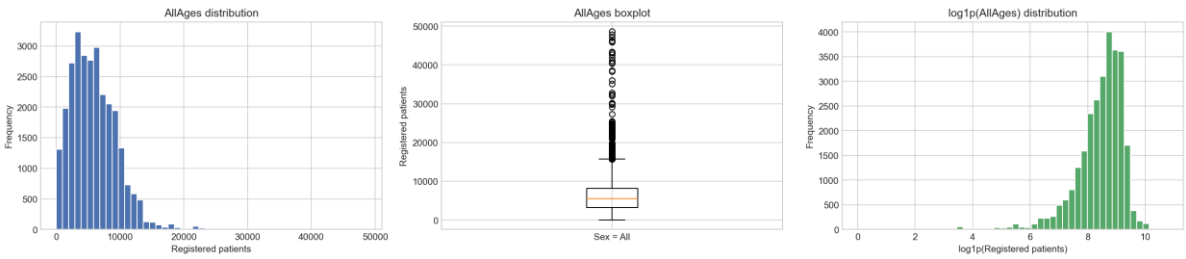

Figure 3. Practice size is strongly right-skewed; a log transformation materially improves symmetry.

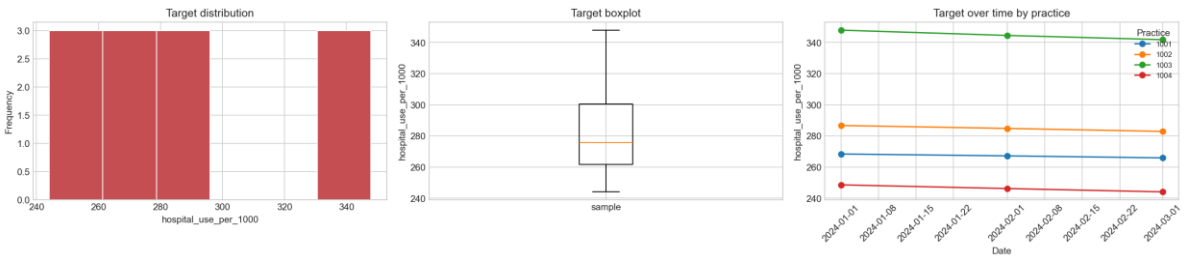

Figure 4. The target-bearing sample is useful for pipeline sanity checks, but it is far too small for substantive inference.

C. Correlation and Dependency Analysis

Three methodological points are especially important. First, age-share features are compositionally dependent because they describe parts of the same practice population. Second, feature-target relationships in the tiny analytical sample are only provisional. Third, multicollinearity is a real modeling risk when totals, shares, and nested geography variables overlap in the information they encode.

The strongest absolute association in the raw share matrix occurs between share_age_25_44 and share_age_65_plus (|r| = 0.669). In the analytical sample, the strongest observed linear link with the target is gp_availability (r = -0.996), but that value should not be overinterpreted because the sample has only 12 rows.

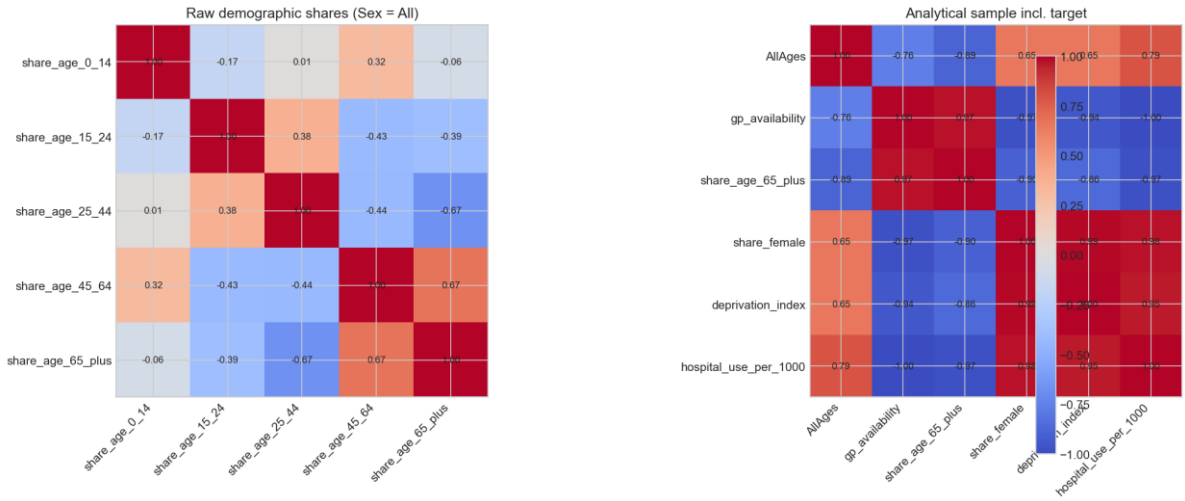

Figure 5. Correlation structure highlights compositional dependence in raw age shares and provisional feature-target links in the sample model table.

C. Correlation and Dependency Analysis — Leakage Risks

Approximate multicollinearity diagnostics point in the same direction: the highest VIF appears for share_age_65_plus (about 697.53), reinforcing the need for parsimonious feature engineering.

• HB and HSCP are hierarchically related, so encoding both naively may introduce quasi-duplication.

• Practice observations repeat over time, so random row-wise splitting can leak practice-specific structure from train to test.

• Raw age counts and derived shares should not be combined indiscriminately because they are algebraically linked through AllAges.

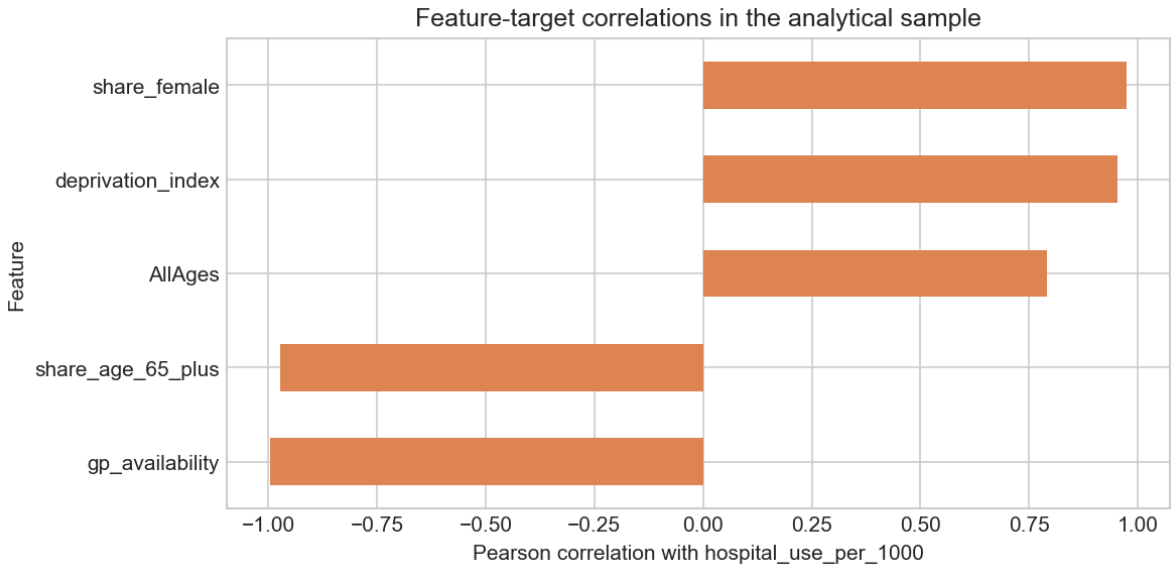

Figure 6. Feature-target correlations in the tiny analytical sample are illustrative only and should not be overinterpreted.

D. Outlier and Noise Assessment / E. Bias and Limitations

Outlier analysis using the IQR rule identifies 375 outlier rows in AllAges, or 1.349% of the aggregate table. These observations are not random corruption; they mainly correspond to structurally large GP practices. The negative Spearman association between practice size and the share of patients aged 65+ is -0.175, suggesting that some outliers belong to a meaningful demographic regime rather than to noise.

The correct engineering decision is therefore to retain large-practice observations while applying transformation or robust scaling to scale-sensitive variables. The broader inferential limits remain important as well: the raw data represent Scottish GP practices only, the observation window includes major system change, and the target-bearing analytical table is not representative of the full thesis setting.

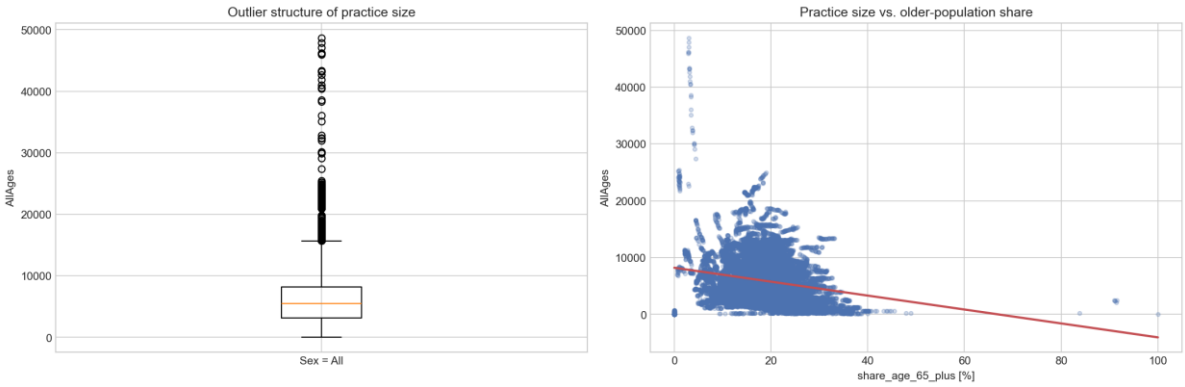

Figure 7. Large practices behave as structurally distinct observations rather than random noise; this supports transformation, not blanket deletion.

F. Implications for Experimental Design

The EDA supports the Assignment 3 hypothesis that demographic structure is a plausible explanatory signal and is therefore compatible with the planned F0 versus F1 comparison. At the same time, it refines the validation strategy needed for Assignment 6.

EDA / continuity finding	**Implication for Assignment 6**
Assignment 3 F0 baseline (practice size + GP availability) remains conceptually coherent.	Keep F0 as the reference condition in later experiments.
Assignment 3 F1 rationale is supported by meaningful demographic and regional variation.	Retain age structure, female share, deprivation, and geography as candidate extensions.
QF columns are almost entirely missing and function as metadata.	Exclude them from quantitative modeling unless a specific quality-flag analysis is justified.
Practice size is strongly right-skewed and contains structurally large practices.	Use log transformation or robust scaling instead of automatic outlier deletion.
Age shares are compositionally dependent.	Prefer a compact derived feature set over many overlapping counts and shares.
Assignment 4 currently uses shuffled row-wise train/test split and shuffled K-fold CV.	Refine Assignment 6 toward time-aware and/or group-aware validation to reduce leakage.
The target-bearing table in the repository is only a tiny reproducibility sample.	Treat target-level findings as provisional until the full real support tables are integrated.

Methodological reflection. This EDA directly justifies what should be excluded (QF metadata), what should be transformed (practice-size related variables), what should be encoded carefully (regional structure), and what should be reconsidered in evaluation design (independence assumptions and leakage risk).